MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **WEEKLY CHAPTER TEST – 03 (LINEAR EQUATIONS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 3: Pair of Linear Equations in Two Variables** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 9(iv).
2. **Section A** comprises 3 Multiple Choice Questions (MCQs) of 1 mark each.
3. **Section B** comprises 1 Assertion-Reason question of 1 mark.
4. **Section C** comprises 2 Very Short Answer (VSAQ) questions of 2 marks each.
5. **Section D** comprises 1 Short Answer (3M) and 1 Long Answer / HOTS (5M) question.
6. **Section E** comprises 1 Applied Case Study with 4 sub-parts of 1 mark each.

SECTION A: Multiple Choice Questions (3 × 1 = 3 Marks)

Q1. If the lines given by $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then the value of k is: [1]

(A) $-5/4$ (B) $2/5$ (C) $15/4$ (D) $3/2$

Q2. The point of intersection of the line represented by $3x - y = 3$ and the y-axis is: [1]

(A) $(0, -3)$ (B) $(0, 3)$ (C) $(1, 0)$ (D) $(-3, 0)$

Q3. The area of the triangle formed by the lines $y = x$, $x = 6$, and $y = 0$ is: [1]

(A) 36 sq. units (B) 18 sq. units (C) 9 sq. units (D) 72 sq. units

SECTION B: Assertion-Reason Question (1 × 1 = 1 Mark)

Q4. **Assertion (A):** The system of equations $2x + y + 9 = 0$ and $x + 3y + 7 = 0$ is consistent with a unique solution.
Reason (R): The system $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is consistent with a unique solution if $a_1b_2 \neq a_2b_1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: Very Short Answer Questions (2 × 2 = 4 Marks)

Q5. Determine the value of k for which the given system of equations has no solution: [2]

$$3x + y = 1 \quad \text{and} \quad (2k - 1)x + (k - 1)y = 2k + 1$$

Q6. Solve the following pair of linear equations using the elimination method: [2]

$$3x + 4y = 10 \quad \text{and} \quad 2x - 2y = 2$$

SECTION D: Short & Long Answer Questions (1 × 3 + 1 × 5 = 8 Marks)

Q7. The sum of a two-digit number and the number obtained by reversing the digits is 66. If the digits of the number differ by 2, find the number. How many such numbers are there? [3]

Q8. Draw the graphs of the linear equations $4x - y - 8 = 0$ and $2x - 3y + 6 = 0$. [5]

(i) Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis.

(ii) Determine the coordinates of the vertices of the triangle formed by these lines and the y-axis.

(iii) Calculate the ratio of the areas of the two triangles.

SECTION E: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q9. Case Study: Social Welfare Coaching Centre Fee Collection

A non-profit educational society conducts remedial mathematics classes for students from diverse backgrounds. To support operations, children from affluent families pay a regular fee of Rs. y per month, while children from economically weaker sections (EWS) receive financial assistance and pay a nominal subsidized fee of Rs. x per month.

In Batch I, there are 20 EWS children and 5 regular children, and the total fee collected is Rs. 9000.

In Batch II, there are 5 EWS children and 25 regular children, and the total fee collected is Rs. 26000.

Based on the above information, answer the following questions:

(i) Represent the fee structure algebraically as a pair of linear equations in x and y. [1]

(ii) Find the monthly fee paid by an EWS child (x). [1]

(iii) Find the monthly fee paid by a regular child (y). [1]

(iv) If Batch III consists of 10 EWS children and 20 regular children, calculate the total collection. [1]

OR

Find the difference in monthly fees paid by a regular child and an EWS child. [1]

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